

Ethics and Data Science

Francisco J. Palmero Moya

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Abstract

The challenge of being an ethical data scientist lies not in understanding ethics but in translating ethical principles into practical applications within the field of data science. While discussions about data ethics are important, the primary issue is the implementation of these ethical principles. The challenge becomes more interesting in predictive optimization, a category of decision-making algorithms that use machine learning (ML) to predict future outcomes of interest about individuals. This document presents a case study based on a rubric of critical questions that are used to deliberate or contest the legitimacy of the specific predictive optimization application proposed by the fictitious company Pizarria.

1 Introduction

In predictive optimization, machine learning is used to make predictions about future outcomes related to individuals, and these predictions are used to inform decisions. While this approach appears appealing by promising objective and data-driven decision-making, it faces various obstacles such as defining what to predict, assembling accurate data, ensuring transparency, and addressing accountability concerns. Wang et al. [10] argue that predictive optimization ultimately fails to deliver on its promises, presenting seven objections to support this claim which are focus of study in the Section 2.

Predictive optimization has generated so much excitement and has been deployed so widely because it has many seemingly attractive characteristics—both in terms of cost and justice. Sansone and Zhu [8] is an example of predictive optimization for welfare fraud early detection who claim that machine learning algorithms can significantly improve predictive accuracy compared to simpler heuristic models or early warning systems currently in use. Predictive optimization promises to improve on each of the two main traditional ways of making decisions: bureaucratic rules and human judgment.

A specific use case of predictive optimization is learning analytics [4, 1], understood as a data-driven decision-making [7]. This viewpoint defines learning analytics as a specific type of analytics where decision-making attempts to improve education and learning [2]. The last few years have witnessed an upsurge in the number of studies using Machine and Deep Learning models to predict vital academic outcomes based on different kinds and sources of student-related data, with the goal of improving the learning process from all perspectives. This has led to the emergence of predictive optimization as a core practice in learning analytics and educational data mining [9].

There is an increasing concern of the interpretability of the analytics from the student experience and what they can tell us about learning [3]. In fact, there is serious debate about the extent to which they are useful as a tool for improving student learning [5]. Some see them as objective measures of student activity, while others argue they lack context. Critics say learning analytics only reveal button clicks, not deeper learning.

In general, predictive optimization is not exempt of inherent objections that cannot be easily fixed using a design or engineering change. These limitations, based on Wang et al. [10] work, are related to specific modeling activities that give rise to them. Thus, since all applications of predictive optimization involve these modeling steps, it suggests that the shortcomings may apply to all of them. Thus, Wang et al. [10] claimed that predictive optimization should be considered presumptively illegitimate, where the need for legitimation is application-dependent. There are several criteria for assessing the legitimacy of predictive optimization that can be used to determine the legitimacy of a decision-making algorithm before it is deployed.

If we consider a more general framework such as Data Science, there has been a great discussion of data ethics, motivated in part by discomfort over “fake news,” targeted advertising, algorithmic bias, and the effect that data products have on individuals and on society. Loukides [6] proposed a checklist to connect principles to practice arguing that they do not leave much room to wiggle compared to, for example, oaths.

The present document takes as legitimacy assessment the rubric based on seven points proposed by Wang et al. [10]. The rubric is based on the following points: good predictions may not lead to good decisions (intervention vs. prediction), it is hard to measure what we truly care about (target-construct mismatch), the training data rarely matches the deployment setting (distribution shifts), social outcomes are not accurately predictable (limits to prediction), with or without machine learning, disparate performance between groups cannot be fixed by algorithmic interventions (disparate performance), providing adequate contestability undercuts putative efficiency benefits (lack of contestability), and predictive optimization does not account for strategic behavior (Goodhart’s law).

2 Case study

In this case study, we imagine that the fictitious company PizarrIA approaches UNED with a proposal to provide a learning analytics service for at-risk students. The company has a system known as risk reporting that relies on artificial intelligence to categorize students as “at risk” based on automated analysis of previous students and their academic outcomes. In this particular case study, the success criterion is defined as each student passing the course (achieving a final grade above 5), and the system must be aware of important dates (start and end of the academic period, exam and assignment submission dates, course enrollment date). Furthermore, they suggest using a set of variables to inform the system for risk prediction, including the number and duration of visits to each activity, participation in activity forums, the number of submissions in each activity, questionnaire attempts, forum interventions, the ratio between recorded and expected daily activity, missed assignments, cumulative days without participation, interactions with peers, spelling errors, library usage, and others. The model will be trained using data from the last three course offerings of the subject. The analysis is then as follows:

Intervention vs. Prediction Predictions often target specific students who appear to be at risk. However, it remains uncertain whether the interventions that the UNED is considering, such as providing advising to individual students, are precisely aligned with the unique needs of each student. For example, professors could provide different options for homework assessment (unique or distributed) to students predicted as at risk. Nevertheless, sometimes individuals may benefit from one option, other times from the other based on their personal preferences. Predictions do not help us understand which interventions would help individual students.

Target-construct mismatch The construct is student being at risk and the target variable is binary variable for passing/no-passing the course. The construct being at risk can be misleading because it may refer to two different situations, dropping out or no-passing the subject. Nevertheless, in both situations the target will be identical since dropping out implies fail. In this sense, construct and target match. However, the binary nature of the target could produce a mismatch when the final grade is lower than five but close to it. In such cases, the professors could decide to pass the student, for example based on its effort during the academic year¹. Nevertheless, if we relax our definition of being at risk from pass/no-pass to be in doubt of pass/no-pass, the mismatch disappears.

Distribution shift The training data for this model come from the last three course offerings of the subject. As the distribution of students attending colleges change over time, the model might not be reliable on new students or those who do not fit the criteria of a typical college student. The last point is clearly relevant in distance learning universities such as UNED. Furthermore, the increasing interest of students in some fields such as artificial intelligence during the last years heavily impacts students distribution attending such lectures over years. In fact, due to the recent developments of artificial intelligence covering new areas such as natural language processing motivates students from these areas to study artificial intelligence this year, unlike past years.

¹Note that this is a mismatch between target and construct, and not a simple wrong prediction because the target (achieve more than five in the final grade) does not match the construct (pass the subject).

Limits to prediction One of the characteristics of predictive optimization is that the prediction target is a future event in an individual's life. Personal situations of each student vary along the academic year and cannot be predicted in most cases. Therefore, the model does not count the student who dropped out for a family emergency, or a reason unrelated to school success itself.

Disparate performance As explained before, students backgrounds may be disparate which could result in disparate performance for different groups. Subjects related to informatics could assign better performance to males than females because of the difference in enrollment proportion².

Contestability There is a lack of information in the description to evaluate this point, however we can introduce one interesting discussions. If the data and model used for making predictions is not available to the students, any inaccuracies are left undetected and unchallenged.

Goodhart's Law Predictive optimization can create unintended incentives for decision subjects to game the system. The UNED could kick out students being at risk so that its graduation rates stay high.

3 Conclusions

In the evolving landscape of data science, the ethical dimension holds paramount significance. As data-driven decision-making becomes increasingly prevalent across fields, ethical considerations must be at the forefront of every data scientist's mind. As we have seen in Section 1, learning analytics is one example where it encompasses techniques that measure, analyze, and report data about participants in a teaching/learning process and their context with the aim of understanding and improving it. However, like many applications of artificial intelligence involving people, there is a heated scientific and philosophical debate about the appropriateness and potential unintended effects of learning analytics. In Section 2 we analyzed a concrete case study showing it presumptively illegitimacy in the context of Wang et al. [10] rubric.

Most concerns presented to predictive optimization are actually closely related to data mining methodology. Thus, target-construct mismatch can be understood as a wrong model hypothesis that does not correspond to reality³; intervention vs. predictions is a simple case of decision-making vs. predictions; distribution shift is a typical case of concept drift⁴; limits to prediction is closely related to model assessment; and disparate performance is similar to non-stratified datasets. Only contestability and Goodhart's Law are more related to deployment, that is, to human behavior. In fact, same argument has been noticed by Wang et al. [10] when they tried to explain why is that these shortcomings cannot be easily fixed using a design or engineering change. The key point here is that we only care about these concerns when the predictions of interest are about humans. If we are trying to infer the Pleiades Open Cluster age, the model could be either valid or simply wrong without need to be legitimate or illegitimate. It is only when we are dealing with humans and decision-making when we must carefully study the consequences of our model. There is when ethics comes into play imposing the set of principles and values that guide human behavior and decision-making, encompassing concepts of right and wrong, fairness, and responsibility.

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²In Spanish universities, the percentage of male-females enrolled in informatics during the academic year 20/21 was 86-14.

³The term reality here could mean, for example, the true generative model.

⁴Concept drift occurs when the underlying patterns or relationships in the data evolve or shift, making models and algorithms trained on historical data less accurate or less relevant for making predictions or decisions in the current or future time periods.

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